

Hybrid Phase and Polarization Sensing Using OIF DSP Frame Pilot Symbols and Training Sequences in Integrated Receivers

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Abstract A hybrid feedforward phase and polarization sensing architecture is proposed for integrated coherent receivers with 100-kHz ECLs, directly utilizing OIF-defined DSP frame pilot symbols and training sequences. By adaptively selecting modalities based on frequency-dependent sensing SNR, we achieve sub-Hz-to-10-MHz vibration detection for fiber-ISAC optical networks. ©2026 The Author(s)

Introduction

Following the establishment of the ITU-T G.681 standard for backscattering fiber-integrated sensing and communication (F-ISAC) [1], hardware-free feedforward sensing in coherent systems has gained considerable momentum toward standardization [2]. Among these techniques, phase- and polarization-based sensing utilizing coherent signals holds substantial promise for sensing-as-a-service (SaaS) [3, 4]. However, practical deployment in existing optical networks faces critical challenges, particularly the phase noise from external-cavity lasers (ECLs) in integrated coherent receivers (ICRs) and the high polarization sensitivity to the environmental activities [5, 6]. To address these challenges, many researchers have devoted considerable effort.

Since the pioneering work for phase sensing was proposed [3], which directly uses the carrier phase recovery (CPR) to track phase, efforts to improve the compatibility with 100-kHz ECLs follow two paths: compensating frequency-drift-induced phase noise (FDPN) [7–9], and mitigating linewidth-induced phase noise (LWPN) [9–11]. However, these methods either require non-standard equipment and algorithms or remain dependent on the vibration frequency, limiting their ability to detect low-frequency disturbances. Conversely, polarization sensing is inherently immune to the laser phase noise. Since polarization states can also be readily extracted from the adaptive equalizer (AEQ) taps and remain highly sensitive to environmental disturbances [4, 6], this approach is widely used in submarine cables for ultra-low-frequency vibration sensing [12–14]. Nevertheless, terrestrial deployments are hindered by severe anthropogenic noise [6]. Hence, more robust metrics with high complexity were proposed for investi-

gating state-of-polarization (SOP) change [6], or polarization information was only used as an assistant for distributed acoustic sensors (DAS) [15]. In summary, developing sensing schemes that are compatible with existing infrastructure and industry implementation agreements (IAs) [16–18] while achieving wideband vibration detection remains a significant challenge.

In this work, we propose a hybrid feedforward phase and polarization sensing architecture utilizing AEQ and CPR, assisted by OIF-defined PS and training sequences (TS). Experimental validation in a 60-GBaud dual-polarization (DP)-16QAM transmission system demonstrates a phase sensitivity of 0.08 rad under 60-kHz vibrations and an SOP angular speed (SOPAS) detection of 0.08 mrad/s. With extra simulations, the proposed architecture is proven suitable for sub-Hz-to-10-MHz wideband sensing. These results have demonstrated its robust performance and excellent compatibility, highlighting a potential for feedforward F-ISAC deployment in intelligent optical networks.

Principle

Inspired by Intel's performance hybrid architecture [19] and the distinct signal-to-noise ratio (SSNR) characteristics of feedforward phase and polarization sensing (Fig. 1), we propose a hybrid sensing architecture that jointly leverages both modalities for low- and high-frequency detection. Then, for coherent systems, due to the

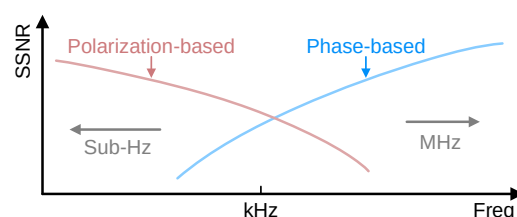


Fig. 1: Concept of hybrid phase and polarization sensing.

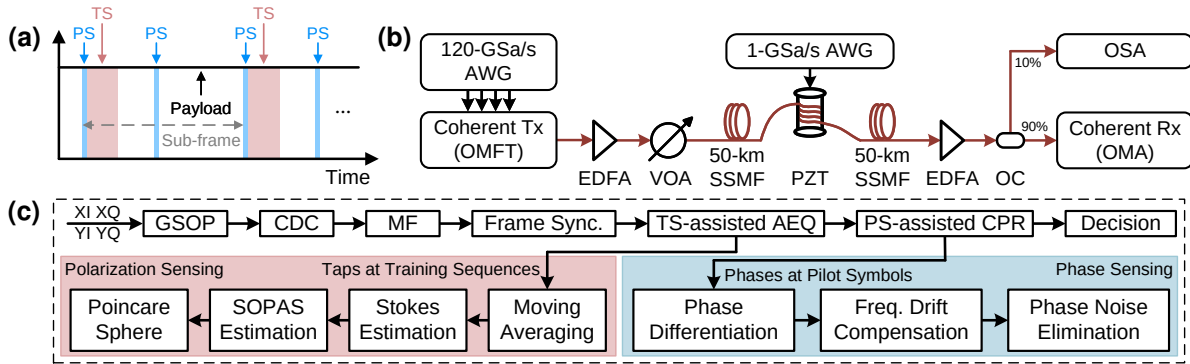


Fig. 2: Block diagram: (a) OIF-defined DSP frame structure, (b) experimental setup, and (c) DSP of coherent systems and TS/PS-based sensing.

low-complexity and non-blind characteristics, the data-assisted algorithms are commonly adopted in commercial deployment [20–23]. As shown in Fig. 2(a), OIF has defined TS and PS [16], assisted with which the AEQ and the CPR can operate, inherently improving the accuracy of feedforward sensing. Building upon these features, the proposed sensing scheme is implemented using the taps of TS-assisted AEQ and the phase tracked by PS-assisted CPR, which is shown in Fig. 2(c).

For phase sensing, it includes 3 steps: tracked phase differentiation, fitting-based FDPN compensation, and averaging-based LWPN elimination. In the CPR, we directly track the phase by subtracting the transmitted signals from the received using PS, whose differential signal can be expressed as:

$$\begin{aligned} \Delta\Phi_{CPR}(k) \\ = \Delta\Phi_{vib}(k) + \Delta\Phi_{FD}(k) + \Delta\Phi_{LW}(k), \end{aligned} \quad (1)$$

where k is the sampling index, corresponding to the PS insertion interval. Φ_{vib} , Φ_{FD} , and Φ_{LW} represent the vibration phase, the FDPN, and the LWPN, respectively. Typically, the frequency drift change is at ~ 1 kHz [9], so, for vibrations with much higher frequency, the differential FDPN can be estimated by polynomial fitting:

$$\Delta\hat{\Phi}_{FD}(k) = \text{Polyfit} \{ \Delta\Phi_{CPR} \} (k), \quad (2)$$

where an order of five is chosen as normal. Then, as illustrated in [9], the differential LWPN $\Delta\Phi_{LW}$ obeys a Gaussian distribution, which means it can be eliminated by N -segment averaging:

$$\hat{\Phi}_{vib}(k) = \sum_{i=1}^k \overline{\Delta\Phi_{CPR,n}(i) - \Delta\hat{\Phi}_{FD,n}(i)}, \quad (3)$$

where n is the index of segments.

Due to the laser noise, the phase sensing can only deal with high-frequency vibrations, while the lower ones will be solved by the polarization sensing. In the AEQ, we use the constant modulus algorithm (CMA), and the radius is set to the outer circle of 16QAM within the TS range. Then, the trained taps matrix can be expressed as:

$$\mathbf{W}(k) = \begin{bmatrix} \vec{w}_{xx} & \vec{w}_{xy} \\ \vec{w}_{yx} & \vec{w}_{yy} \end{bmatrix}, \quad (4)$$

where $\vec{w}_{ij}$ is from j input to i output tributaries. Then, the Stokes parameters can be calculated:

$$\begin{aligned} s_0 &\propto |w_{xx}|^2 + |w_{xy}|^2, \\ S_1 &= [|w_{xx}|^2 - |w_{xy}|^2] / s_0, \\ S_2 &= 2 \cdot \Re \{ w_{xx}^* w_{xy} \} / s_0, \\ S_3 &= 2 \cdot \Im \{ w_{xx}^* w_{xy} \} / s_0, \end{aligned} \quad (5)$$

where $w_{ij} = \sum \vec{w}_{ij}$ is the sum of taps in time domain. To reduce the channel noise, a moving averaging operation can be applied to these taps. The SOPAS of Stokes vector is expressed using:

$$\frac{\Delta SOP}{\Delta t} = \frac{2}{\Delta t} \arcsin \left[\frac{1}{2} \sqrt{\sum (\Delta S_i)^2} \right], \quad (6)$$

where Δt is the TS repeating time interval i.e., $\Delta t = R_s \times L_{sub-frame}$, supposing R_s is the symbol rate and $L_{sub-frame}$ is the number of symbols in a sub-frame. Finally, the low-frequency vibration-induced polarization change can be analyzed by the SOPAS and the Poincaré sphere.

Experimental Results

Figure 2(b) illustrates the experimental setup. We transmitted 60-GBaud DP-16QAM data. While this work was based on the OIF400ZR-defined digital signal processing (DSP) frame [16], the underlying principles are equally applicable to forthcoming OIF800ZR/OIF1600ZR or other IAs [17, 18]. It was generated by a 120-GSa/s arbitrary waveform generator (AWG, Keysight 8194A) and modulated by an optical multi-format transmitter (OMFT, ID Photonics OMFT-C-00-FA). An integrated digital coherent receiver with oscilloscope (optical modulation analyzer, OMA, Keysight N4391B) was used to receive the optical signal. The linewidths of ECLs were both ~ 100 kHz. The optical signal-to-noise ratio (OSNR) was controlled by adjusting the launch power with an Erbium-doped fiber amplifier (EDFA) and a variable optical attenuator (VOA), monitored by an optical spectrum analyzer (OSA), and set under a bit error rate (BER) of $\sim 1.5 \times 10^{-2}$. Between the 50-km standard single-mode fibers (SSMFs), we used a 1-GSa/s-AWG-driven piezoelectric transducer (PZT) to emulate vibrations. The offline DSP is shown in Fig. 2(c).

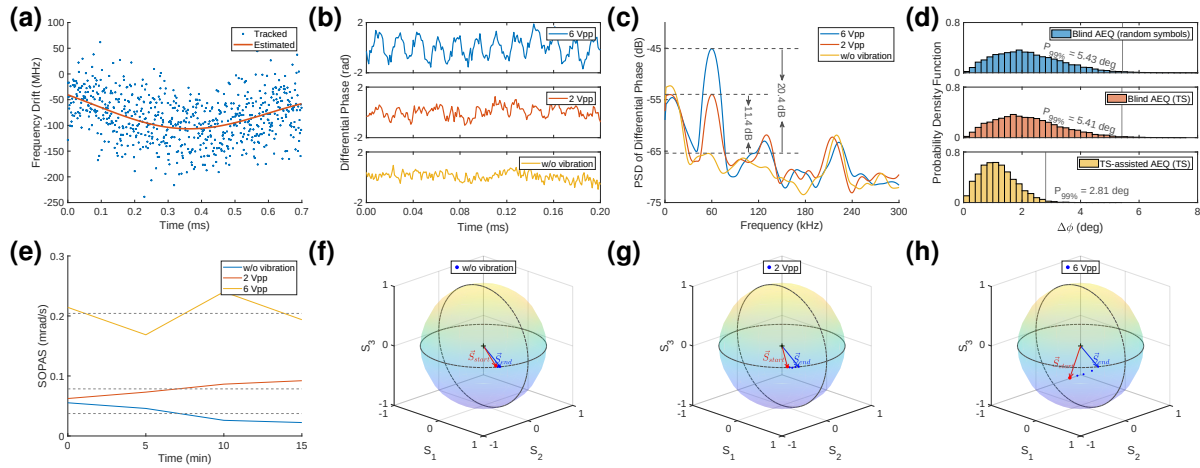


Fig. 3: Experimental results. Phase sensing: (a) tracked and estimated frequency drift, and (b-c) waveforms and PSDs of extracted differential phase under different-amplitude vibrations. Polarization sensing: (d) histograms of angular deviation using blind/TS-assisted AEQ; (e) estimated SOPAS and (f-h) tracked Poincaré spheres under different-amplitude vibrations.

For phase sensing, the high-frequency sensing performance was investigated, so we applied 60-kHz and $0 \sim 6\text{-}V_{pp}$ peak-to-peak voltages to the PZT. The frequency components calculated by $\Delta\Phi_{CPR}(k)/2\pi$ and $\Delta\Phi_{FD}(k)/2\pi$ are shown in Fig. 3(a), which indicates that the frequency drift was well estimated by polynomial fitting. Figure 3(b) gave the extracted differential phase waveforms. When vibrations emulated by 6 and $2\text{-}V_{pp}$ were applied, $\sim 2.7\text{-}$ and $\sim 0.9\text{-}$ rad peak-to-peak values were clearly observed, compared with no-vibration situations. In Fig. 3(c), the power spectrum densities (PSDs) of differential phase indicated the SSNR reached 20.4 and 11.4 dB. The 9.0-dB difference was consistent with the theoretical calculated by $(V_{pp,1}^2/V_{pp,2}^2)\text{dB}$ i.e., $\sim 9.5\text{ dB}$, showing excellent linearity. The noise floor was -65.4 dB, meaning the sensitivity reached 0.08 rad.

In Figs. 3(d)-(h), we considered polarization sensing. The SOP angular deviation can illustrate the noise floor [6], so we investigated them using different AEQs and different part of taps to calculating Stokes parameters. As a non-constantly modulated 16QAM, a blind algorithm cascading CMA and radius-directed equalization (RDE), and the TS-assisted CMA were taken into consideration. The results in Fig. 3(d) indicated that, with the same data amount, the deviation of TS-assisted CMA was more concentrated in the lower area, and the 99% of the angular deviation reached 2.81° , which was much better than the other two ($\sim 5.4^\circ$). These results could indicate that, by using TS-assisted AEQ, the noise floor can be reduced. Then, we performed polarization sensing, with the sampling interval set as 5 minutes. Figure 3(e) illustrates the estimated SOPAS. It showed the averages were ~ 0.21 and ~ 0.08 mrad/s for 6- and $2\text{-}V_{pp}$ -emulated vibrations, respectively, also showing great linearity. Compared with the results without vibration, the $2\text{-}V_{pp}$ -emulated one can still be easily detected. Finally, the trace length in

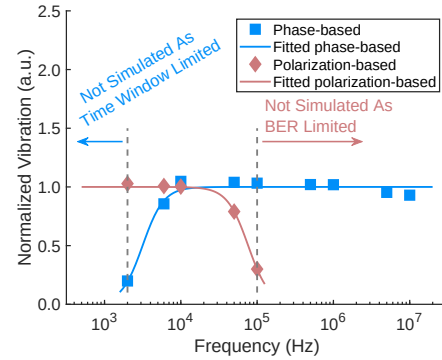


Fig. 4: Simulation results: feasibility verification of hybrid sensing for wideband-frequency vibration detection.

Poincaré spheres of Figs. 3(f)-(h) also demonstrated about three times for $6\text{-}V_{pp}$ than $2\text{-}V_{pp}$.

We also performed a simulation to verify the feasibility of hybrid sensing. Different-frequency vibration with fixed 10-rad phase and the corresponding SOPAS (~ 10 times the value of frequency [24]) were applied, and we calculated the normalized vibration (the extracted divided by the applied). As illustrated in Fig. 4, an indicator value closer to 1 signifies superior modality linearity across the frequencies. These results indicate that the phase is well-suited for vibration sensing with frequencies higher than $\sim 10\text{ kHz}$, while the polarization is matched for lower ones, which fully validates the motivation behind the hybrid sensing architecture. These results verify its sensing capability for wideband vibrations from sub-Hz to 10 MHz.

Conclusions

A feedforward hybrid sensing architecture utilizing OIF DSP frame TS-assisted AEQ and PS-assisted CPR is proposed. Experiments revealed 0.08-rad phase sensitivity and successful 0.08-mrad/s SOPAS detection. Simulations well demonstrated the feasibility of hybrid sensing for sub-Hz-to-10-MHz wideband scenarios. These validated its robust performance and compatibility.

Acknowledgements

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